SEONVIN CHO

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Research Interests

Reinforcement Learning · Sequential Decision Making · Generative Models

Education

Hanyang University, Seoul, Republic of Korea Sep. 2024 – Present
Integrated M.S./Ph.D. Student in Electronic Engineering
Advisor: Prof. Songnam Hong
GPA: 4.45 / 4.50

Hanyang University, Seoul, Republic of Korea Mar. 2020 – Aug. 2024
B.S. in Electronic Engineering
Summa Cum Laude, Early Graduation
GPA: 4.26 / 4.50

Research

Recent Work

Role-Adaptive Policy Optimization for Offline Reinforcement Learning 2026
Seonvin Cho, Soohyun Choi, Songnam Hong† · *Preprint* · †Corresponding author
[arXiv]

- Proposed Role-Adaptive Policy Optimization (RAPO), which learns policy-update coefficients according to whether actions are used for critic bootstrapping or for execution.
- Separated bootstrap and execution actors in TD3+BC and adapted only the inverse temperature for IQL, improving both base algorithms on D4RL locomotion and AntMaze.

Is One Step Enough for Offline Policy Improvement? 2026
Soohyun Choi*, **Seonvin Cho***, Songnam Hong† · *Preprint* · *Equal contribution · †Corresponding author
[Code] [arXiv]

- Studied multi-step proximal policy improvement that re-centers each stage on the preceding policy, asking when a single improvement step is enough.
- Distinguished subdivision of a fixed total horizon from additional refinement, with experiments on TD3+BC and IQL-based policy extraction in D4RL locomotion.

PathBridger: Subgoal Bridges for Offline Goal-Conditioned Reinforcement Learning 2026
Soohyun Choi*, **Seonvin Cho***, Songnam Hong† · *Preprint* · *Equal contribution · †Corresponding author
[Code] [arXiv]

- Developed a hierarchical offline goal-conditioned RL framework that connects subgoal selection to short-horizon execution through explicit state-space bridges.
- Combined generative endpoint proposals, transitive value-based selection, and inverse-dynamics decoding to produce executable short-horizon action chunks.

Federated Learning under Noisy Labels 2023
Undergraduate Capstone Project
[Code] [Technical Report]

- Analyzed loss-value distributions in federated learning to identify unreliable samples under label noise.
- Received the Best Award (Department) and Excellence Award (College of Engineering).

Ongoing Research

Extensions of PathBridger Present

- **Skill Learning and Composition:** Investigating how to learn reusable skills from offline trajectories and compose them for long-horizon goal-conditioned tasks.
- **Latent-Space Bridge Planning:** Exploring learned representations and geometrically structured latent spaces for goal-conditioned bridge generation and short-horizon execution.

Honors & Awards

BK Research Assistantship (BK-RA), Hanyang University — Spring & Fall 2026
IEEE Antennas and Propagation Society Undergraduate Summer Research Scholarship — 2024
Excellence Paper Award, KIEES [Code] — Winter & Fall 2024
Best & Excellence Awards, Federated Learning Capstone Project — 2023
Excellence Award, Embedded Creative Robot Challenge — 2022
Merit-Based & Academic Scholarships, Hanyang University — 2021 – 2024